

SPC Case Studies - Real-World Process Control Failures and Wins

4 case studies: what went wrong, root cause, lesson learned

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

CS-SPC01 - Bore Diameter: Cpk 1.82 at PPAP, then 1.18 at Audit

Product:	CNC machined bore, CC characteristic, Ppk 1.84 at PPAP approval
Problem:	IATF audit 8 months after SOP: auditor calculated Cpk from recent control chart data = 1.18. Major NC raised.
Root Cause:	CNC tool wore down over time. SPC chart showed 3 Nelson Rule 3 trends before audit, but operators did not respond. No out-of-control log maintained.
Lesson:	Cpk is not a one-time PPAP number. Ongoing monitoring is required. Nelson rules must be trained and enforced - not just displayed.
Corrective Action:	Tool change frequency reduced from 500 to 300 pieces. Operator reaction plan retraining. Monthly Cpk review added to Management Review inputs.

CS-SPC02 - Control Limits Set Equal to Spec Limits

Product:	Weld tensile test, CC characteristic, drawing spec 450-550 MPa
Problem:	SPC chart found at audit with UCL=550, LCL=450. Control chart showed all points within limits for 6 months - perfectly flat. No real data variation.
Root Cause:	Quality engineer set control limits from the spec limits, not from process data. Process variation was much smaller - a real UCL would be ~520 MPa.
Lesson:	Control limits must come from process data. Spec limits on the chart are dashed reference lines only. Setting UCL=USL destroys the chart's ability to detect process shifts.
Corrective Action:	Recalculated control limits from 25 subgroups of actual data. UCL=512 MPa, LCL=488 MPa. Retraining for all quality engineers on control limit calculation.

CS-SPC03 - Operator Tampering - Rule 4 Violation

Product:	Assembly press force, SC characteristic, target 20 kN +/- 2 kN
Problem:	SPC chart showed Rule 4 pattern (alternating up-down) for 3 weeks. Cpk calculated as 1.45.
Root Cause:	Operator was adjusting press pressure setting after every reading - even when process was in control. This is 'tampering' - it increased variation, not decreased it.
Lesson:	Nelson Rule 4 (14 alternating points) is a signature of over-adjustment / tampering. The process has its own natural variation. Adjusting a stable process makes it worse.
Corrective Action:	Retraining: 'If the chart shows no special cause, do not adjust.' Press set-up procedure locked - adjustment requires QA approval. Cpk improved to 1.78 after tampering stopped.

CS-SPC04 - Proactive Cpk Improvement Through SPC

Product:	Hardness, CC, specification 58-62 HRC
Problem:	Cpk at PPAP: 1.68 (just above 1.67 minimum). Customer required quarterly Cpk reviews. After 6 months, Cpk trending down toward 1.55.
Root Cause:	Not a failure - proactive detection. Furnace temperature uniformity degrading gradually. SPC trend caught it 3 months before Cpk would breach 1.33.
Lesson:	SPC is most valuable when used proactively. The trend chart prevented a customer escape. Quarterly Cpk review enabled early action.
Corrective Action:	Furnace elements replaced. Cpk improved to 1.93. Cost of furnace maintenance: Rs.1.2L. Avoided cost of customer return, recall, IATF NC: estimated Rs.35L+.